👩‍🏫 Staff Guide · Chemistry · Unit 1 Lightning (Level 3, tester)

Level 3 = resource-room students who can work close to the gen ed lesson. No slide decks: students follow Mr. Brittle's slides with Level 3 guided notes and readings.

1. How the pieces line up

Gen ed lesson	Level 3 piece
L1.1-L1.6 (initial model, smaller scales, sparks, particle model, cause of static)	Part 1A: reading + guided notes (table, drawing, short answers) + quick check
L1.3 Sparks / L1.5 / L1.8 balloon activities	Static Electricity Stations: Level 3 lab sheet with data table + CER
L1.7 Coulomb's Law (k, scientific notation, sig figs)	Part 1B notes: proportional reasoning (×2 charge → ×2 force; ×2 distance → ÷4 force) + scientific notation practice. No k-constant calculations or sig figs
L1.8 Neutral Interactions, L1.9 Review	Part 1B reading, drawing, quick check
L1.10-L1.12 (energy flow, the strike, conductors & insulators)	Part 1C reading + notes (strike table, conductor sort)
L1.13 No Swimming + L1.13.5 You Light Up My Life	Conductivity Lab: Level 3 lab sheet (data table, CER, amps-to-charges math)
L1.14 Practice test + Unit test	Part 1C quick check + Level 3 unit game; the paper Unit Test is 🚧 (not in the export)

2. Supports built in

- Every reading and notes page has an online copy with a 🔊 Read aloud button (highlights as it reads)
- Key-word box at the top of each notes page; worked math example before every math problem
- Fewer filled-in parts than Level 2: short answers, a table, a drawing
- No slide decks: follow Mr. Brittle's slides

3. What to grade

- Guided notes stand in for the matching L1.x worksheets (same points when complete)
- Quick checks: score, then scale to the class points
- Lab sheets: same points as the class lab
- Modified grading only as the IEP allows. Check with the case manager.

4. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm say “check with Mr. Brittle”.

⚡ Charges & Static: Guided Notes

Chemistry · Unit 1A · Level 3 (closer to the class work) · goes with L1.1-L1.6 (Initial Model, Smaller Scales, Sparks, Particle Model of Static, Cause of Static)

Name: _____

Use the class slides and your reading. Key words:

atom *nucleus* *proton* *neutron* *electron* *charge* *neutral* *static charge* *correlation* *causation*

A. Lightning and evidence (L1.1, L1.2)

1. Lightning is a giant electric _____ between a cloud and the ground.
2. Rain and lightning happen together. This is a _____.
3. Why does a correlation NOT prove that rain causes lightning?

_____

_____

B. The atom (L1.5)

4. Fill in the table.

Particle	Charge	Where is it?	Can it move to another object?
Proton			
Neutron			
Electron			

5. An atom with equal numbers of protons and electrons is _____.
6. Draw a neutral atom with 3 protons, 3 neutrons and 3 electrons. Label each particle and its charge.

C. Static charge and sparks (L1.3, L1.6)

7. Rubbing a balloon on hair moves _____ to the balloon.

8. The balloon becomes _____ and the hair becomes _____.

9. Explain why the hair is now positive. Use the words electron and proton.

10. What is a spark? Use the word charge.

⚡ Charges & Static: Reading

Chemistry · Unit 1A · Level 3 (closer to the class work) · goes with L1.1-L1.6 (Initial Model, Smaller Scales, Sparks, Particle Model of Static, Cause of Static)

Name: _____

What Causes Lightning?

Lightning is a giant electric spark. It can jump from a cloud to the ground or between clouds. It heats the air to about 30,000 °C, about five times hotter than the surface of the sun. The hot air expands so fast that it makes a boom: thunder.

Scientists noticed that clouds that make a lot of rain also make a lot of lightning. This is a correlation: the two things happen together. A correlation does not prove that one thing causes the other. To find the cause, we need a model at a smaller scale.

All matter is made of atoms. In the center of each atom is the nucleus. It holds protons, which have a positive (+) charge, and neutrons, which have no charge. Around the nucleus are electrons, which have a negative (−) charge. A neutral atom has the same number of protons and electrons.

Protons are locked in the nucleus, but electrons can move from one object to another. When you rub a balloon on your hair, electrons move from your hair to the balloon. The balloon now has extra electrons, so it is negative. Your hair is missing electrons, so it is positive. This build-up of charge is called static charge.

Inside a storm cloud, ice crystals and soft hail (graupel) bump into each other as air moves up and down. Each bump moves electrons, like rubbing a balloon. The bottom of the cloud becomes negative and the top becomes positive. When enough charge builds up, electrons jump through the air. That jump is lightning.

Check your reading

1. What is the difference between a correlation and a cause?

2. Why do electrons, and not protons, move when you rub a balloon?

3. How is a storm cloud like a rubbed balloon?

🎈 Static Electricity Stations: Lab Sheet

Chemistry · Unit 1A · Level 3 · goes with L1.3 Sparks, L1.5 Particle Model of Static, L1.8 Neutral Interactions

Name: _____

Question: How do charged objects interact with other charged and neutral objects?

Materials

2 balloons, paper scraps, tinsel, thin stream of water, string or tape

⚠️ Safety

- Tell staff about latex allergies.
- Keep charged balloons away from computers and outlets.
- Wipe up water spills right away.

Prediction

Predict: will a charged balloon attract or repel neutral paper? Explain using electrons.

_____

_____

Procedure

1. Rub a balloon on hair or wool for 10 seconds.
2. Station 1: hold it 2 cm above paper scraps. Record.
3. Station 2: hold it near tinsel. Record.
4. Station 3: hold it near a thin stream of water. Record.
5. Station 4: charge two balloons, hang them 5 cm apart. Record.
6. Move each object farther away and record if the effect gets weaker.

Data

Station	Attract or repel?	Neutral or charged object?	What happened when farther away?
1 Paper scraps			
2 Tinsel			
3 Water stream			
4 Two balloons			

Analysis (Claim, Evidence, Reasoning)

1. Claim: How does a charged balloon interact with neutral objects?

2. Evidence: Which stations support your claim?

3. Reasoning: Explain using electrons, polarized, and distance.

⚡ Charges & Static: Quick Check

Chemistry · Unit 1A · Level 3 (closer to the class work) · goes with L1.1-L1.6 (Initial Model, Smaller Scales, Sparks, Particle Model of Static, Cause of Static)

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Which particle has a negative charge?

A. Proton | B. Neutron | C. Electron

2. Where are protons found?

A. In the nucleus | B. Outside the nucleus | C. Between atoms

3. Rain and lightning happen together. This is evidence of…

A. A proven cause | B. A neutral charge | C. A correlation

4. A balloon rubbed on hair becomes negative because it…

A. Gains electrons | B. Loses protons | C. Gains neutrons

5. An object has 10 protons and 12 electrons. Is it positive, negative or neutral? Explain.

6. Explain how charge builds up in a storm cloud.

⚡ Charges & Static: Guided Notes · ANSWER KEY

Chemistry · Unit 1A · Level 3 (closer to the class work) · goes with L1.1-L1.6 (Initial Model, Smaller Scales, Sparks, Particle Model of Static, Cause of Static)

Name: _____

Use the class slides and your reading. Key words:

atom · *nucleus* · *proton* · *neutron* · *electron* · *charge* · *neutral* · *static charge* · *correlation* · *causation*

A. Lightning and evidence (L1.1, L1.2)

1. Lightning is a giant electric **spark** between a cloud and the ground.
2. Rain and lightning happen together. This is a **correlation**.
3. Why does a correlation NOT prove that rain causes lightning?
Two things can happen together because of a third cause; we need evidence of the mechanism.

B. The atom (L1.5)

4. Fill in the table.

Particle	Charge	Where is it?	Can it move to another object?
Proton	**+ (positive)**	**nucleus**	**no**
Neutron	**0 (none)**	**nucleus**	**no**
Electron	**− (negative)**	**outside the nucleus**	**yes**

5. An atom with equal numbers of protons and electrons is **neutral**.
6. Draw a neutral atom with 3 protons, 3 neutrons and 3 electrons. Label each particle and its charge.
3 + in the nucleus with 3 neutral neutrons, 3 − around the outside, all labeled.

C. Static charge and sparks (L1.3, L1.6)

7. Rubbing a balloon on hair moves **electrons** to the balloon.
8. The balloon becomes **negative** and the hair becomes **positive**.
9. Explain why the hair is now positive. Use the words electron and proton.
The hair lost electrons but kept all its protons, so it has more + than −.
10. What is a spark? Use the word charge.
Built-up charge (electrons) jumping across a gap of air.

🔑 *Key written by our team from Mr. Brittle's worksheets and slides, please check.*

🎈 Static Electricity Stations: Lab Sheet · ANSWER KEY

Chemistry · Unit 1A · Level 3 · goes with L1.3 Sparks, L1.5 Particle Model of Static, L1.8 Neutral Interactions

Name: _____

Question: How do charged objects interact with other charged and neutral objects?

Materials

2 balloons, paper scraps, tinsel, thin stream of water, string or tape

⚠️ Safety

- Tell staff about latex allergies.
- Keep charged balloons away from computers and outlets.
- Wipe up water spills right away.

Prediction

Predict: will a charged balloon attract or repel neutral paper? Explain using electrons.
Attract: the balloon pushes the paper's electrons away, so the near side is + (polarized).

Procedure

1. Rub a balloon on hair or wool for 10 seconds.
2. Station 1: hold it 2 cm above paper scraps. Record.
3. Station 2: hold it near tinsel. Record.
4. Station 3: hold it near a thin stream of water. Record.
5. Station 4: charge two balloons, hang them 5 cm apart. Record.
6. Move each object farther away and record if the effect gets weaker.

Data

Station	Attract or repel?	Neutral or charged object?	What happened when farther away?
1 Paper scraps	**attract**	**neutral**	**weaker**
2 Tinsel	**attract**	**neutral**	**weaker**
3 Water stream	**attract**	**neutral**	**weaker**
4 Two balloons	**repel**	**charged (both −)**	**weaker**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: How does a charged balloon interact with neutral objects?
It attracts them.

2. Evidence: Which stations support your claim?
Paper, tinsel and water all moved toward the balloon.

3. Reasoning: Explain using electrons, polarized, and distance.
The balloon's − charge pushes electrons to the far side; the near side is +, so it is pulled; the force is weaker farther away.

🔑 *Key written by our team from Mr. Brittle's lab sheets, please check.*

⚡ Charges & Static: Quick Check · ANSWER KEY

Chemistry · Unit 1A · Level 3 (closer to the class work) · goes with L1.1-L1.6 (Initial Model, Smaller Scales, Sparks, Particle Model of Static, Cause of Static)

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Which particle has a negative charge?

A. Proton | B. Neutron | **C. Electron**

2. Where are protons found?

A. In the nucleus | B. Outside the nucleus | C. Between atoms

3. Rain and lightning happen together. This is evidence of…

A. A proven cause | B. A neutral charge | **C. A correlation**

4. A balloon rubbed on hair becomes negative because it…

A. Gains electrons | B. Loses protons | C. Gains neutrons

5. An object has 10 protons and 12 electrons. Is it positive, negative or neutral? Explain.

Negative: it has 2 more electrons (−) than protons (+).

6. Explain how charge builds up in a storm cloud.

Ice and graupel collide as air moves; collisions move electrons, so the bottom becomes negative and the top positive.

🔑 *Reworded from his L1.9 review and L1.14 practice test, please check.*